Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Memory Production Paper (6) - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Tue, Sep 29, 2026 at 9:29 AM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

You are right. Calling my random search "the more informative test" was wrong: the supremum of $h_n/\|\Delta_V\|$ is approached near Flat_n and Dirichlet draws do not land there, so the search never visited the region where a violation would occur. Random maxima understate the supremum, as your 0.999 along the worst linear direction shows.
I didn't check it properly when suggested by AI assistance.

So I ran the test that does examine that region. At a flat family W, with $L(E)(t,s,r) = E_{ts} - E_{tr}W_{rs} - W_{tr}E_{rs}$,
the admissible space $A = \{E : \mathbf{1}^\top E = 0,\; E(j{\leftarrow}i)p_i = 0\}$ and T the tangent space of Flat_n,
the first-order ratio is $\|P_{T^\perp}E\|/\|L(E)\|$ with supremum $1/\sigma_{\min}$.
Over forty-two flat families, $n \in \{2,3,4\}$, $m \in \{3,4,5\}$: $\sigma_{\min} \geq 1$ in every one, minimum 1.000013,
so the first-order supremum is 0.999987 (your 1.00003 over 175 families, by another route).

One addition. You state the evidence as the smallest nonzero singular value, and that restriction is an assumption rather than
a convenience: a direction $E \in A$ with $L(E) = 0$ and $P_{T^\perp}E \neq 0$ would make the deficit vanish to first order while
the distance to Flat_n did not, leaving the bound vacuous. So $\ker L \cap A \subseteq T$ needs checking.
It holds: across the forty-two families the relative norm of $P_{T^\perp}(\ker L \cap A)$ is of order 10^{-16}.
With that discharged your evidence is complete; still first-order evidence, and the paper is right to print only the question.

Revision 2.2. The content is correct and is what my verification had established.
The placement is the part worth acknowledging: this manuscript follows the memory rank paper in the series, so the citation is backward
and creates none of the circularity the reverse link would have.

I will check the wording against the deposited text of 2.2; nothing above depends on it.

Best,
Ezequiel

[Quoted text hidden]